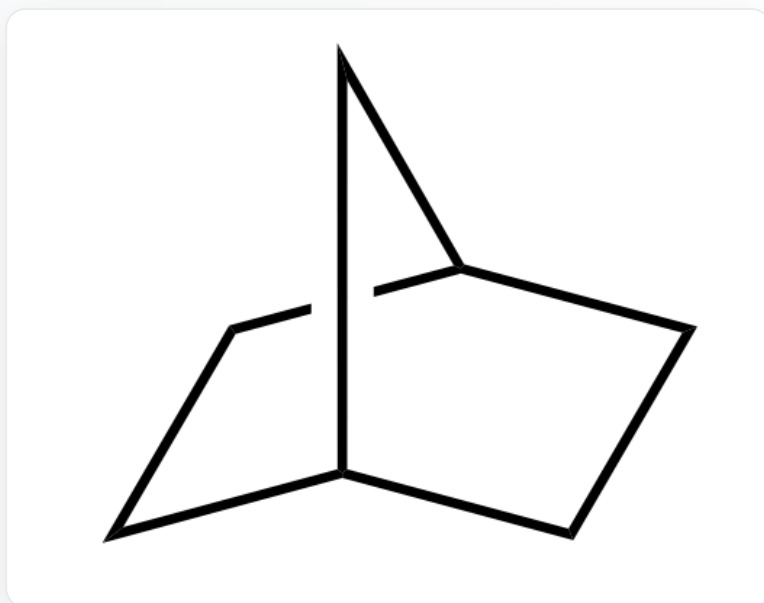


Question

Norbornene was first treated with borane and then hydrogen peroxide under basic conditions to obtain compound A. Compound A was further oxidized with chromium trioxide and reacted with p-toluenesulfonyl hydrazide to yield compound B. Compound B was then reacted with ethylene oxide and trimethylsilyl chloride under the action of n-butyllithium to obtain compound C. Compound C was reacted with m-chloroperbenzoic acid to yield compound D. Compound D was treated under acidic conditions to obtain compound E.

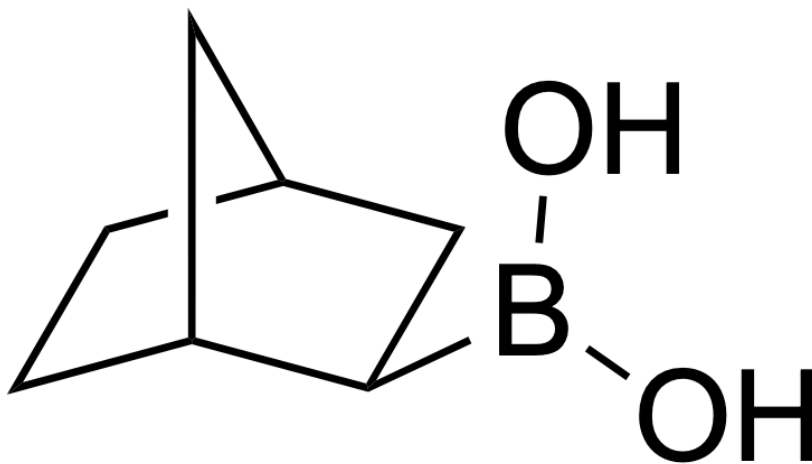
Please provide the structures of all compounds and select the matching option from the following choices.

A. Compound A is



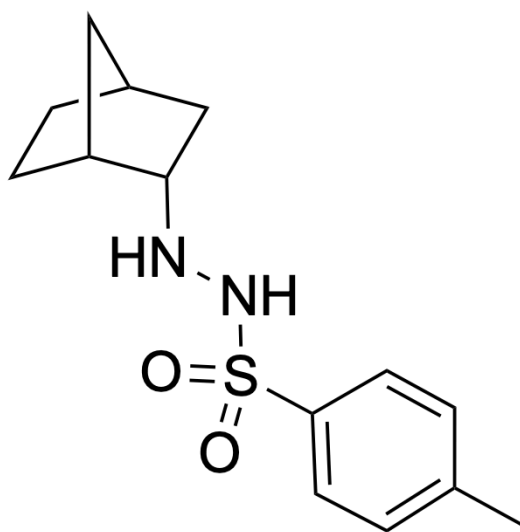
[C@H]12CC[C@@H](C2)CC1

B. Compound A is



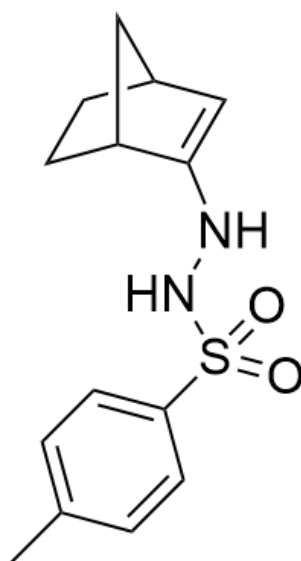
OB(O)[C@H]1[C@@H]2CC[C@@H](C2)C1

C. Compound B is



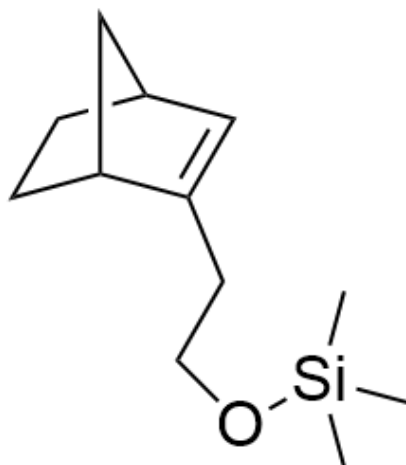
O=S(NN[C@@H]1[C@@H]2CC[C@@H](C2)C1)(C3=CC=C(C)C=C3)=O

D. Compound B is



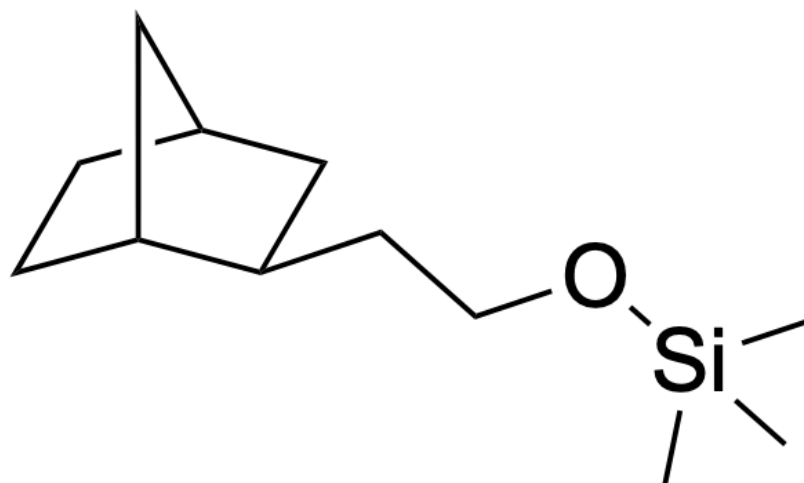
O=S(NNC1=C[C@H]2CC[C@@H]1C2)(C3=CC=C(C)C=C3)=O

E. Compound C is



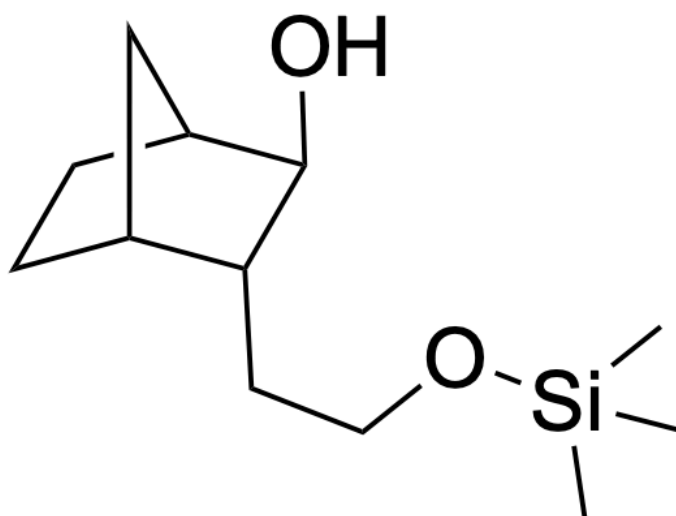
C[Si](C)(C)OCCC1=C[C@H]2CC[C@@H]1C2

F. Compound C is



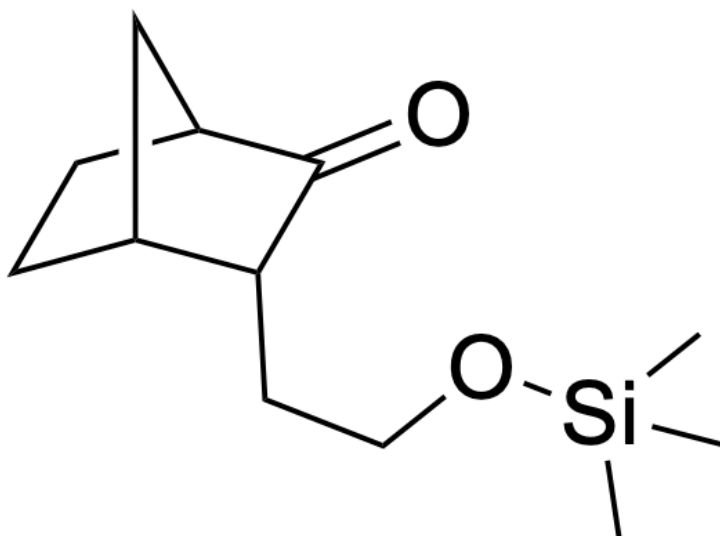
C[Si](C)(C)OCC[C@H]1[C@@H]2CC[C@@H](C2)C1

G. Compound D is



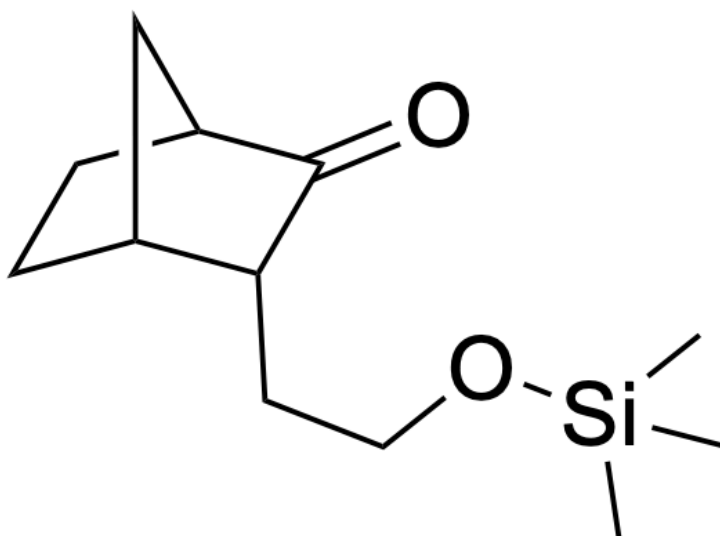
C[Si](C)(OCC[C@H]1[C@@H]2CC[C@H]([C@H]1O)C2)C

H. Compound D is



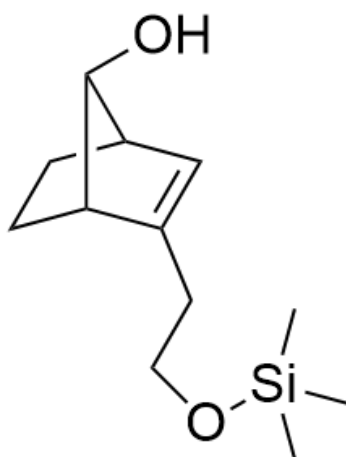
C[Si](C)(C)OCC[C@H]1[C@@H]2CC[C@H](C2)C1=O

I. Compound E is



C[Si](C)(C)OCC[C@H]1[C@@H]2CC[C@H](C2)C1=O

J. Compound E is



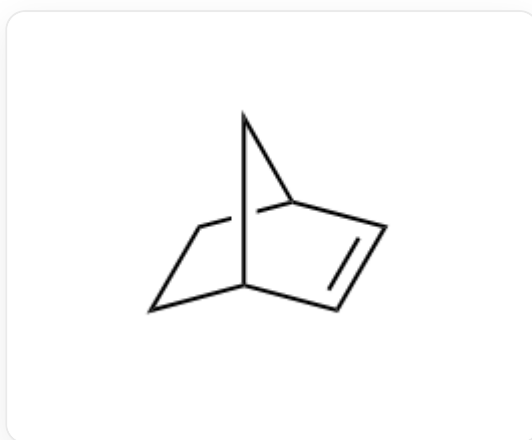
C[Si](C)(C)OCCC1=C[C@H]2CC[C@@H]1[C@H]2O

Answer

Correct Answer: **E**

Detailed Explanation

First, norbornene is



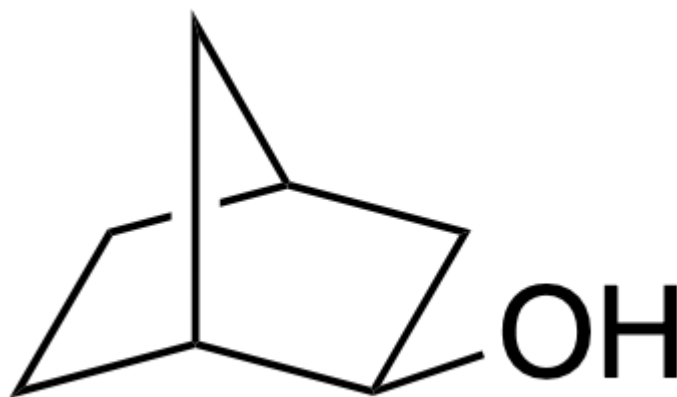
[C@H]12CC[C@@H](C2)C=C1

CHECKPOINT

0.2 PTS

Norbornene is [C@H]12CC[C@@H](C2)C=C1

Borane can undergo an addition reaction with alkenes, and under hydrogen peroxide conditions, rearrangement occurs, ultimately yielding the product of apparent water addition to the alkene. Therefore, compound A is



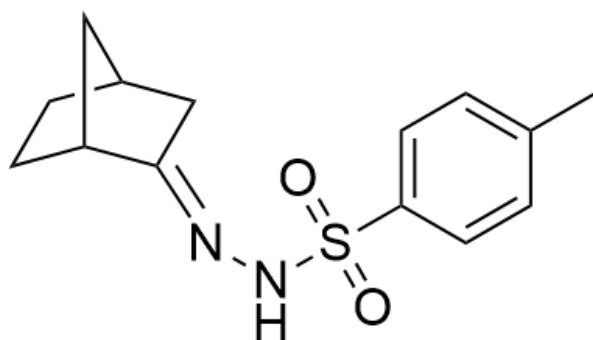
O[C@H]1[C@@H]2CC[C@@H](C2)C1

CHECKPOINT

0.5 PTS

Compound A is O[C@H]1[C@@H]2CC[C@@H](C2)C1

Chromium trioxide can oxidize hydroxyl groups to aldehyde groups, and the aldehyde can condense with p-toluenesulfonylhydrazine to form an imine. Thus, compound B is



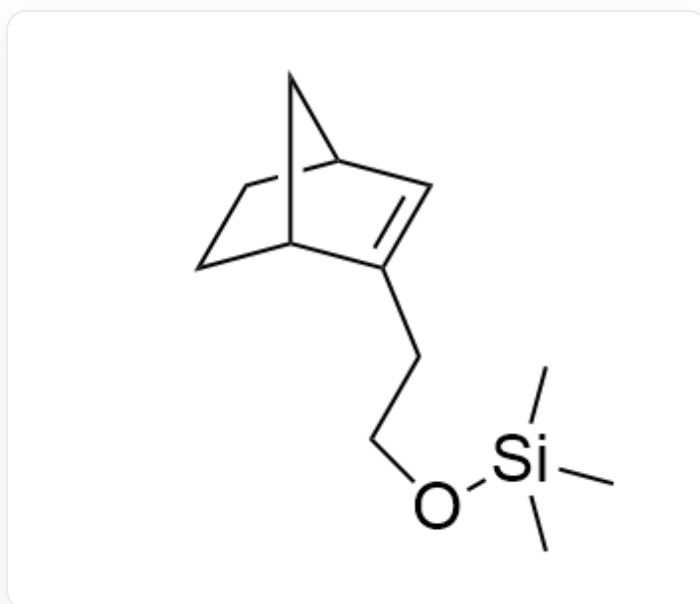
O=S(=O)(N1C=CC2CC[C@@H](C2)C1)C1=CC=C(C=C1)C

CHECKPOINT

1 PTS

Compound B is O=S(N/N=C1[C@@H]2CC[C@@H](C2)C/1)(C3=CC=C(C=C3)C)=O

The proton on nitrogen is acidic and can be removed by reaction with a base, resonating to carbon and undergoing nucleophilic addition with ethylene oxide. The resulting oxyanion reacts with trimethylchlorosilane due to the hard-soft acid-base principle. At this stage, the adjacent secondary C-H bond is acidic and can be deprotonated, eliminating the azo and p-toluenesulfonyl groups. Consequently, compound C is



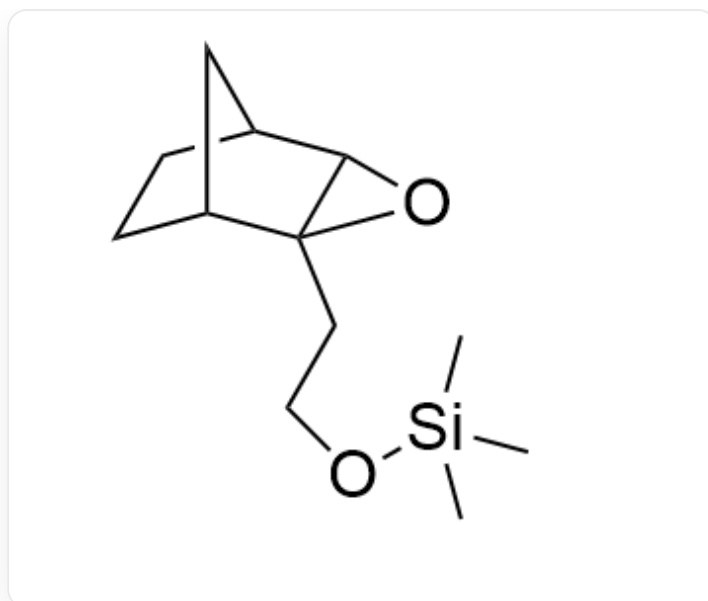
C[Si](C)(C)OCCC1=C[C@H]2CC[C@@H]1C2

CHECKPOINT

1.5 PTS

Compound C is C[Si](C)(C)OCCC1=C[C@H]2CC[C@@H]1C2

Next, meta-chloroperoxybenzoic acid can convert the alkene group into an epoxide, leading to compound D as



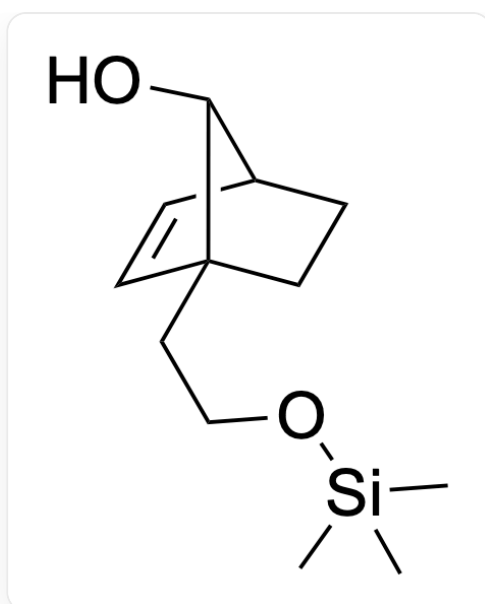
C[Si](C)(OCC[C@]12[C@@H]3CC[C@H]([C@H]1O2)C3)C

CHECKPOINT

0.5 PTS

Compound D is C[Si](C)(OCC[C@]12[C@@H]3CC[C@H]([C@H]1O2)C3)C

Finally, under acidic conditions, the epoxide group protonates to form a carbocation, which undergoes rearrangement via the norbornene skeleton and elimination, resulting in compound E as



O[C@@H]1[C@@H]2C=C[C@]1(CCO[Si](C)(C)C)CC2

CHECKPOINT

2 PTS

Compound E is O[C@@H]1[C@@H]2C=C[C@]1(CCO[Si](C)(C)C)CC2